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Inventor(s)	Ivri; Yehuda

Vibration systems and methods

Abstract

In one arrangement, a vibration system includes a vibratable plate, a support member surrounding the vibratable plate, and a vibration-inducing member surrounding the support member. The vibration-inducing member is configured to radially expand and contract against the support member so as to produce axial vibration of the vibratable plate. In another arrangement, the vibratable plate has an outer circumference; a tubular member is concentrically disposed about the outer circumference of the plate, and an annular vibration-inducing member is concentrically disposed about the outer circumference of the tubular member. The vibration-inducing member is preferably a piezoelectric ring that is radially expandable and contractable against the wall of the tubular member to cause the plate to vibrate in the axial direction.

Inventors:	Ivri; Yehuda (Newport Beach, CA)
Applicant:	Stamford Devices Ltd (Galway, IE)
Family ID:	37452540
Assignee:	Stamford Devices Ltd (Galway, IE)
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Primary Examiner: Boecker; Joseph D.

Attorney, Agent or Firm: Bookoff McAndrews, PLLC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/570,177, filed on Sep. 13, 2019, which is a continuation of U.S. patent application Ser. No. 16/240,516, filed Jan. 4, 2019, which is a continuation of U.S. patent application Ser. No. 14/816,975, filed Aug. 3, 2015, which is a continuation of U.S. patent application Ser. No. 11/920,805, filed on Nov. 19, 2007, now U.S. Pat. No. 9,108,211, which is a 371 filing of PCT/US2006/014654, filed on Apr. 17, 2006, which claims the benefit of U.S. Provisional Application No. 60/684,720, filed on May 25, 2005, all of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

(1) This invention relates generally to a vibration system that efficiently transfers radial vibration from a vibration-inducing member to produce axial vibration in a vibratable member through a support member that filters out undesirable vibration. In preferred embodiments, a piezoelectric

transducer impacts ultrasonic oscillation to a vibratable plate, particularly a vibrating aperture (orifice) plate of an aerosol generator device, wherein the vibrating plate is perforated with holes and is operable in a fluid medium. The invention may also be useful in the field of ultrasonic wave transmission in a fluid medium including, but not limited to, underwater sonar, depth sonar and obstacle detecting sonar.

(2) Devices wherein a circular aperture plate is vibrated using a piezoelectric transducer in the form of a ring are well-known in the art. For example, in Maehara, U.S. Pat. No. 4,605,167, a vibratable plate having at least one nozzle opening therein is secured to a rigid housing and a piezoelectric ring is secured to the vibratable plate for inducing therein a displacement to discharge a small quantity of liquid through the nozzle opening. In other devices described in the art, a circular vibratable plate is directly bonded to the piezoelectric ring and covers the central opening in the piezoelectric ring; for example, as described in Toda, U.S. Pat. No. 5,297,734. As other examples, Humberstone et al., U.S. Pat. No. 5,518,179, Ross et al., U.S. Pat. No. 5,261,601 and Davison et al., U.S. Pat. No. 6,062,212 describe vibrating devices wherein a circular vibratable plate is mounted over the central opening of a thin annular member, or “washer”, which is bonded to a piezoelectric ring. When actuated, the piezoelectric ring vibrates radially to cause the washer to operate in a “bending mode” that vibrates the vibratable plate in the axial direction. When these types of devices are manufactured the components must be carefully aligned concentrically with each other when the vibratable plate is attached to the washer and the washer is positioned over the opening in the piezoelectric ring.

(3) Installing an ultrasonic transducer directly onto a rigid frame or housing, such as taught by Maehara, is also problematic. The reason is that the entire body of the transducer vibrates, with some portions vibrating at a small amplitude and some portions vibrating at an amplified amplitude. When the transducer is installed on the rigid fixture, the oscillation amplitude is reduced. Another problem with prior art arrangements is that the piezoelectric ring naturally vibrates in 3 directions (i.e., X, Y and Z axes) and transmits such vibration to the vibrating plate. The vibrations that are transmitted to the plate are superimposed and the contribution of the axial vibration may be canceled or partially canceled out. It is therefore desirable to filter out the undesirable vibration of the piezoelectric element and to use only vibration in a single direction, e.g., the axial direction. Attempts to address this problem have included mounting the vibratable plate on resilient retainer members, or “fingers”, e.g., as described in Martens III et al., U.S. Pat. No. 6,450,419; supporting a bimorph-type transducer on rubber O-rings, e.g., as described in Ross et al., U.S. Pat. No. 5,261,601 and Humberstone et al., U.S. Pat. No. 5,518,179; and holding the piezoelectric ring in place with a grommet, e.g., as described in Helf et al., U.S. Pat. No. 6,293,474 and Tomkins et al., U.S. Pat. No. 6,382,522.

(4) The above-described devices translate the radial vibration of a piezoelectric ring to axial vibration of a perforate plate to disperse a liquid as an aerosol. In another type of device, the piezoelectric transducer may be tubular and may expand and contract axially (in the direction of the central axis of the tube) to move a perforate plate. For example, see Newcombe et al., U.S. Pat. No. 5,838,350.

(5) Generally, in a piezoelectric transducer that operates in a fluid medium, such as those used in aerosolizers, there is a need to separate the piezoelectric element from the vibratable element so that the vibratable element can be submerged in liquid and the piezoelectric element can be electrically insulated from the liquid. In some cases, such insulation may be provided by encasing the piezoelectric element with elastomer material. Such material, while providing electrical insulation, also has energy-absorbing characteristics that dampen the oscillation amplitude of the piezoelectric transducer and therefore has an adverse effect on the efficiency of the device.

(6) There are currently a wide variety of aerosolizers and nebulizers. Of particular interest are those which vibrate an aperture plate or other element to produce the aerosol. Examples of some of these aerosolizers are described in U.S. Pat. Nos. 5,169,740, 5,938,117, 6,540,154, 5,586,550, 5,750,647,

6,467,476, 6,014,970, 6,755,189, 6,814,071, 6,554,201, 6,732,944, 6,615,824, 6,845,770, and 6,851,626, each of which is incorporated herein by reference in its entirety. One issue with aerosolizers (especially those used for medical applications using highly corrosive liquids) is contamination and corrosion of parts. Certain parts of the aerosolizer that are exposed to liquids may need to be washed or disposed of in order to keep the aerosolizer in good working order. Many of these parts are difficult to clean and, since piezoelectric transducers and their associated electronics can be relatively expensive, making them disposable may not be economically feasible. Therefore, it may be desirable to make certain aerosolizer components removable or replaceable.

BRIEF SUMMARY OF THE INVENTION

(7) One or more embodiments of the present invention are directed to a vibration system that efficiently transfers radial vibration from a vibration-inducing member to produce axial vibration in a vibratable member through a support member that filters out undesirable vibration. In one particular arrangement, the vibration-inducing member may be an annular piezoelectric transducer having a central opening, the vibratable member may be a thin circular plate and the support member may have a circular cross-section for holding the circular plate; however, the components may also possibly have other shapes. In other arrangements, the vibration-inducing member is separable from the other structures so that it may be reused for other applications. In addition, other arrangements of the present invention effectively isolate the vibration-inducing member from contact with liquids and is therefore particularly adapted for operation in a fluid medium, e.g., in an aerosolizer environment. Still further, the present invention provides a method for self-aligning the components of a vibration system comprising a piezoelectric ring and a circular vibratable plate so as make manufacture of the system simple and inexpensive.

(8) In one or more embodiments, the vibration system of the present invention comprises a thin circular vibratable plate, a tubular member holding the vibratable plate and a piezoelectric ring coupled to the tubular member. The vibratable plate may be concentrically disposed within the lumen of a thin-walled tubular member; and a piezoelectric ring may be concentrically positioned about the outer circumference of the tubular member at the location of the vibratable plate. The piezoelectric ring is expandable and contractable in the radial direction, which in turn causes the walls of the tubular member to expand and contract in the radial direction. This movement of the tubular member walls expands and contracts the outer circumference of the plate causing its middle region to oscillate (i.e., vibrate) in the axial direction. Since the outer circumference of the vibratable plate of the present invention is positioned within the central opening of the piezoelectric ring in alignment with its central plane, in contrast to prior systems wherein a surface of the vibratable plate (or concentric washer around the plate) is secured at the surface of the piezoelectric ring across the central opening of the piezoelectric ring, the radial load produced by the piezoelectric ring is more symmetrically applied to the vibratable plate of the present invention.

(9) In one embodiment, the vibratable plate may be an aperture plate that includes a plurality of tapered apertures and is preferably dome-shaped. The aperture plate may be coupled to a mounting structure disposed within the lumen of the tubular member that holds the aperture plate perpendicular to the central axis of the tubular member.

(10) In another embodiment, the tubular member may be fabricated from a corrosive-resistant metallic material, e.g., a palladium/nickel alloy or stainless steel, and have flexible thin walls, e.g., less than 0.5 mm in thickness. In other embodiments, the tubular member may comprise a plastic material, and may include at least one resilient segment, e.g., an elastomer, disposed therein that allows the tubular member to be compressed and expanded by the piezoelectric ring. In another embodiment, the piezoelectric ring may be removable from the tubular member and re-used for other applications. For example, the tubular member may be tapered to form a "taper lock" wherein the piezoelectric ring is press-fitted with the tubular member. In a still further embodiment, the piezoelectric ring may be permanently bonded to the tubular member to form an integral unit.

(11) In one embodiment, a tubular member containing an aperture plate may be operably coupled

to a reservoir of liquid, or the reservoir may be an integral part of the tubular member, so that liquid is supplied to the aperture plate within the tubular member. In this way, when the aperture plate is vibrated in accordance with the invention, liquid droplets are ejected from the aperture plate in the form of an aerosol. Optionally, a ring may be disposed about the outer periphery of the piezoelectric ring to assist in vibrating the aperture plate at its resonant frequency. In embodiments wherein the tubular member and the piezoelectric ring are not integrated with the reservoir, O-rings or other seals may be provided between the reservoir and the tubular member to serve as liquid-tight seals that prevent contamination of the piezoelectric ring, and also provide damping the piezoelectric ring and the aerosolizer housing, thereby increasing the efficiency of the system. In another embodiment, the tubular member may be “press-fitted” (i.e., form an “interference fit”) with a discharge opening of the reservoir.

(12) In one embodiment of the invention, the reservoir and the tubular member containing the aperture plate may be integrated as a single unit, and the piezoelectric ring may be slid over the reservoir of the aerosolizer to form a press-fit with the tubular member. In this way, the piezoelectric ring may be removed without potential contamination from substances within the reservoir or tubular member, or from aerosol produced by the aperture plate. In another embodiment, the piezoelectric ring and the tubular member may be bonded together as an integral unit.

(13) The vibration system of the present invention may be incorporated into a variety of products and may be connected to power supplies, electronics to vibrate the vibration-inducing member, and the like. In one embodiment, the vibration system of the present invention is connected to a power supply using a first wire that makes electrical contact with a first surface of the piezoelectric ring and a second wire that makes electrical contact with a second surface of the vibration-inducing member. These wires may be located in grooves surrounding the outer surface of the housing in which the vibration system is located so that they are isolated from liquids in the reservoir and within the tubular member of the vibration system.

(14) Examples of products that may employ the vibration system of the present invention include ventilators, continuous positive airway pressure (CPAP) systems, hand-held nebulizers and the like, as well as devices that utilize ultrasonic wave transmission in a fluid medium, such as, for example, various sonar devices. As one example, a ventilator circuit may comprise a length of tubing, and the vibration system of the present invention may be operably coupled to the tubing to introduce aerosol generated by the vibration system into the ventilator circuit. As another example, an aerosolizer may be constructed of a housing having a mouthpiece and the vibration system of the invention may be disposed in the housing so that liquid droplets produced by the vibrating aperture plate are ejected through the mouthpiece and into the respiratory system of the user.

(15) One embodiment of the invention provides an exemplary method for making a vibration system comprising the steps of inserting a vibratable plate into a support structure that surrounds the plate; surrounding the support structure including vibratable plate with a vibration-inducing member that is configured to expand and contract radially; and actuating the vibration-inducing member to produce radial expansion and contraction against the support member to cause axial vibration of the vibratable plate.

(16) In one particular embodiment, a method of making a vibration system comprises the steps of providing a tubular member with a lengthwise lumen, securing a circular vibratable plate within the lumen so that the vibratable plate is perpendicular to the central axis of the tubular member; providing a piezoelectric ring having a center opening; positioning the tubular member within the center opening of the piezoelectric ring so that outer circumference of the tubular member is in contact with the inner circumference of the opening and the vibration-inducing member surrounds the vibratable plate within the tubular member; and securing the vibration-inducing member to the tubular member.

(17) In another embodiment, a method of vibrating a plate comprises the steps of inserting a

vibratable plate in a support structure that surrounds the entire periphery of the plate, inserting the support structure including vibratable plate into the central opening of a vibration-inducing member that is configured to expand and contract radially, and actuating the vibration-inducing member to produce radial expansion and contraction against the support member that causes axial vibration of the vibratable plate.

(18) In another embodiment, a method of treating a patient is provided, which comprises the steps of providing a vibration system comprising a circular vibratable aperture plate having an outer circumference, a tubular member concentrically disposed about the outer circumference of the vibratable plate, wherein the tubular member has an outer circumference, and an annular vibration-inducing member concentrically disposed about the outer circumference of the tubular member, wherein the vibration-inducing member is radially expandable and contractable to cause the aperture plate to vibrate in the axial direction; supplying a liquid medicament to the aperture plate via the tubular member; actuating the vibration-inducing member to vibrate the aperture plate and aerosolize the liquid medicament; and supplying the aerosol to a patient's respiratory system.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1*a* and 1*b* are exploded perspective views of a vibration system of the invention.
- (2) FIG. 2 is a partial cross-sectional view of the assembled vibration system of FIGS. 1*a* and 1*b*.
- (3) FIGS. 3*a* and 3*b* are cross-sectional side views of the vibration system of FIG. 2.
- (4) FIG. 4*a* is a cross-sectional side view of one embodiment of an aerosolizer according to the invention.
- (5) FIG. 4*b* is a partial cross-sectional view of the aerosolizer shown in FIG. 4*a*.
- (6) FIG. 4*c* is a perspective view of the aerosolizer shown in FIG. 4*a*.
- (7) FIG. 4*d* is a cross-sectional side view of another embodiment of an aerosolizer according to the invention.
- (8) FIG. 5*a* is a perspective view of another embodiment of a vibration system according to the invention.
- (9) FIG. 5*b* is a partial cross-sectional view of the vibration system shown in FIG. 5*a*.
- (10) FIG. 6 is a perspective view of another embodiment of a vibration system according to the invention.
- (11) FIG. 7*a* is a perspective view of another embodiment of an aerosolizer according to the invention.
- (12) FIG. 7*b* is a cross-sectional side view of the aerosolizer of FIG. 7*a*.
- (13) FIG. 8 is a cross-sectional side view of another embodiment of an aerosolizer system according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

- (14) All publications, patents and patent applications cited herein, whether supra or infra, are hereby incorporated by reference in their entirety to the same extent as if each individual publication, patent or patent application was specifically and individually indicated to be incorporated by reference.
- (15) It must be noted that, as used in this specification and the appended claims, the singular forms “a”, “an” and “the” include the plural unless the content clearly dictates otherwise.
- (16) Reference herein to “one embodiment”, “one version” or “one aspect” shall include one or more such embodiments, versions or aspects, unless otherwise clear from the context.
- (17) In one or more embodiments, the vibration system of the present invention comprises a vibratable plate, a support member surrounding the vibratable plate, and a vibration-inducing member surrounding the support member, wherein the vibration-inducing member is configured to

radially expand and contract against the support member so as to produce axial vibration of the vibratable plate. The following detailed description is directed to one preferred embodiment of the invention wherein the vibratable plate is circular, the support member has a circular cross-section, e.g., a tubular member (cylindrical or tapered), into which the circular vibratable plate is disposed, and the vibration-inducing member is an annular disc having a central opening, i.e., a piezoelectric ring, into which the support member is disposed. However, it should be understood that the invention is not limited to this embodiment.

(18) The tubular member may be manufactured from a corrosion-resistant metal, for example, stainless steel (preferably grades 316, 303 or 416), titanium, or a C-276 chrome/nickel alloy (e.g., Hastelloy® C-276). The tubular member preferably has relatively thin walls that can be effectively deflected by the piezoelectric ring. In one embodiment, the thickness of the walls of the tubular member is in the range of 0.1 mm to 0.5 mm, preferably about 0.25 mm. In one embodiment, the tubular member may have a shelf structure disposed around its inner surface to which the periphery of the vibratable plate may be bonded so that it extends across the internal lumen of the tubular member perpendicular to its central axis.

(19) Various piezoelectric rings known in the art may be suitable for use as the annular vibration-inducing member of the present invention. In one embodiment, the piezoelectric ring may comprise any material exhibiting piezoelectric properties, for example, a piezoelectric ceramic material such as lead zirconate titanate (PZT) or lead metaniobate (PN) and may take the shape of a disc of substantially constant thickness with a central hole. Such piezoelectric rings are commercially available, e.g., from American Piezo Ceramics, Inc. (APC), Mackeyville, Pennsylvania, and from Morgan Electro Ceramics (MEC), Fairfield, New Jersey. The piezoelectric ring may be supplied with an alternating electric current at the selected frequency from a power source; for example, the piezoelectric ring may be electrically connected by wires to a controller that contains the electronics necessary to control the vibration of the piezoelectric ring.

(20) In accordance with the invention, the tubular member is positioned within the center opening of the piezoelectric ring. When actuated by the alternating electrical fields from the controller, the piezoelectric ring expands and contracts in the radial direction against the walls of tubular member in the vicinity of the vibratable plate. This movement of the tubular member walls expands and contracts the periphery of the vibratable plate, thereby forcing the center of the vibratable plate to oscillate in the axial direction, i.e., to move up and down along the central axis of the tubular member. Although the piezoelectric ring may also vibrate in the axial direction and may create a transverse surface wave, only the radial vibration can be transmitted to the vibratable plate by the tubular member. In this way, the superposition of conflicting vibration modes is eliminated and efficient translation of electrical energy to mechanical movement is accomplished. The practice of the present invention also allows the vibration system to be installed directly to a rigid body, such as the frame or housing of an aerosolizer, nebulizer or other device, without having the vibration transfer to the entire body. This is mainly because the ends of the tubular member do not vibrate and therefore may be used to install the vibration system to the rigid body.

(21) The invention may be particularly useful when the tubular member is employed to hold an annular aperture plate or other structure having a plurality of apertures. When a liquid is applied to one side of the aperture plate through the tubular member and the piezoelectric ring is actuated, the aperture plate oscillates in a manner that causes liquid droplets to be ejected from the apertures. The resultant aerosol may then be dispensed out the open end of the tubular member.

(22) A particularly useful type of aperture plate is one having tapered apertures that taper from the surface contacting the liquid to the surface where the droplets are ejected. Also, in some embodiments, the aperture plate may be domed shaped, although the invention is not limited to only such aperture plates. Preferred aperture plates may have a thickness in the range of 20 to 100 microns. Examples of piezoelectric materials and aperture plates that may be used with the invention are described in U.S. Pat. Nos. 6,235,177 and 5,758,637, incorporated herein by

reference. In another embodiment, the piezoelectric ring may be vibrated at a frequency in the range from about 20 KHz to about 500 KHz, for example, about 128 KHz. In another preferred embodiment, the droplets may have a size suitable for use in pharmaceuticals, for example, in the range from about 3 micrometers (μm) to about 6 μm , and the liquid may be aerosolized at a rate in the range from about 5-20 microliters/second.

(23) Referring now to FIGS. 1a, 1b and 2, one embodiment of the present invention will be described. Vibration system **10** comprises vibratable plate **101**, tubular member **102** and piezoelectric ring **103**. Tubular member **102** has an outer circumference **104** and an inner circumference **105**, which together define a relatively thin cylindrical wall, preferably having a thickness in the range from about 0.1 mm to 0.5 mm. The hollow center (lumen) of tubular member **102** terminates in openings **106** and **107** at opposing ends thereof. Mounting structure **111** comprises a circular ridge that projects perpendicularly from inner circumference **105** into the lumen of tubular member **102** at a location, preferably a central location, between openings **106** and **107**. Piezoelectric ring **103** comprises an annular disc of piezoelectric material having a center hole **108** with a circumference **112** approximately equal to the outer circumference **104** of tubular member **102**. Vibratable plate **101** comprises circular outer flange **109** surrounding a thin circular vibratable center portion **110**.

(24) In one method of making vibration system **10**, metallic tubular member **102** may first be provided with mounting structure **111** by bonding a ridge of metal around inner circumference **105** at a location equidistant from ends **106** and **107**. Vibratable plate **101** may then be concentrically disposed within the lumen of tubular member **102** with the lower surface of circular flange **109** positioned over the upper surface of mounting structure **111** and with the outer periphery of vibratable plate **101** abutting inner circumference **105**. Outer flange **109** of vibratable plate **101** may be secured onto mounting structure **111** using a suitable joining procedure, e.g., a metallurgical process such as brazing, welding, soldering or the like, or a chemical bonding process such as adhesive bonding.

(25) In one preferred embodiment, a brazing ring of a suitable corrosion-resistant brazing filler material, e.g., a mixture of 70% gold and 30% copper, may be placed between the upper surface of mounting structure **111** and outer flange **109** of vibratable plate **101**. The entire assembly of tubular member **102**, vibratable plate **101** and brazing ring may be held in place by a weight placed on top of vibratable plate **101**. The assembly may be placed in an oven and heated to a temperature sufficient to melt the brazing **1** and permanently join the surfaces together in a conventional brazing procedure. In another embodiment, vibratable plate **101** may be soldered onto mounting structure **111** using soldering materials, such as a tin/lead soldering material; however, this method may not be suitable if the assembly is to be exposed to acidic pharmaceutical preparations. In another embodiment, vibratable plate **101** may be secured onto mounting structure **111** by ultrasonic or laser welding.

(26) Once vibratable plate **101** is secured across the lumen of tubular member **102**, tubular member **102** may be positioned within center hole **108** of piezoelectric ring **103**. In one embodiment, tubular member **102** may be placed in a fixture that holds tubular member **102** upright, and piezoelectric ring **103** may be slid lengthwise down tubular member **102** until piezoelectric ring **103** surrounds the outer circumference **104** at a location directly corresponding to the location of mounting structure **111** and vibratable plate **101** on inner circumference **105** of tubular member **102**. Outer circumference **104** of tubular member **102** and circumference **112** of center hole **108** in piezoelectric ring **103** may then be bonded together, e.g., by depositing a suitable liquid adhesive around the juncture of circumference **104** and circumference **112** and curing the adhesive, e.g., with UV light. The adhesive used should be capable of efficiently transferring vibration from the piezoelectric ring **103** to tubular member **102**. Although ideally the adhesive would have the modulus of elasticity ("Young's Modulus") of the piezoelectric ring, i.e., about 60 GPa (Giga Pascal), to achieve the ultimate transfer of vibration, this is not possible for any adhesive. Most

structural adhesives (such as epoxy) have a modulus of elasticity of plastic material, which may be about 2 GPa, and should be suitable for the present invention if cured to approximately that stiffness. As examples of suitable adhesives, mention may be made of various epoxy and anaerobic adhesives, such as commercially available UV-cured epoxy adhesives sold under the trademark Loctite.

(27) As previously described, piezoelectric ring **103** is configured to radially expand and contract when alternating electric fields are communicated to it via electric lines. For example, as illustrated in FIG. **3a**, piezoelectric ring **103** contracts radially towards its center opening (direction D) when actuated by a first electric field. This radial contraction causes piezoelectric ring **103** to push inward along outer circumference **104** of tubular member **102** in the vicinity of mounting structure **111** and thereby pinch the wall of tubular member **102**. The constriction of tubular member **102** causes flange **109** to also constrict radially and, as a result, the center portion **110** of vibratable plate **101** moves axially in direction A. When actuated by a second electric field, as shown in FIG. **3b**, piezoelectric ring **103** expands radially away from its center opening (direction D'), thereby releasing the inward pressure along circumference **104** of tubular member **102**. This release of pressure allows flange **109** to expand radially, which causes center portion **110** of aperture plate **101** to move axially in direction A' to its original position. Continually alternating the electric fields produces an oscillation (vibration) of center portion **110** between the positions shown in FIGS. **3a** and **3b**.

(28) As previously mentioned, the vibration system of the present invention is particularly useful for aerosolizing liquids. FIGS. **4a**, **4b** and **4c** illustrate an aerosolization system (referred to herein as "aerosolizer **40**") in accordance with embodiments of the present invention. The same reference numbers are used in each of the Figures to refer to the same component. Referring now to FIG. **4a**, aerosolizer **40** comprises upper housing **401**, detachable lower housing **405** and vibration system **10** (e.g., see FIG. **2**). Vibration system **10** comprises tubular member **102**, piezoelectric ring **103** and aperture plate **101**. Upper housing **401** comprises reservoir **402** configured to hold a volume of liquid, e.g., a liquid medicament, and a conical portion **403** at the lower end of reservoir **402** terminating in discharge tube **404** defined by cylindrical walls **406**. Engagement tube **407** defined by cylindrical walls **408** of upper housing **401** is concentrically disposed around and completely encompasses discharge tube **404**.

(29) Vibration system **10** is adapted to be detachably engaged with upper housing **401**, with the upper section of tubular member **102** of vibration system **10** (i.e., that section of tubular member **102** above piezoelectric ring **103**) being configured to be press fit within discharge tube **404** and with piezoelectric ring **103** of vibration system **10** being configured to be press fit with engagement tube **407**. When assembled, the upper section of tubular member **102** of piezoelectric ring **103** is fully encompassed by discharge tube **404** and the top surface of piezoelectric ring **103** abuts the lower end of discharge tube **404**. This press fit mating of tubular member **102** and discharge tube **404** forms a liquid-tight seal that prevents liquid discharged from reservoir **402** into discharge tube **404** from coming in contact with piezoelectric ring **103**.

(30) Lower housing **405** comprises receiving tube **411** defined by cylindrical walls **412**, annular flange **413** concentrically disposed around the base of receiving tube **411** and aerosol chamber **414** defined by cylindrical walls **415**. Lower housing **405** may be adapted to be detachably engaged with vibration system **10** and upper housing **401**, with the lower section of tubular member **102** of vibration system **10** (i.e., that section of tubular member **102** below piezoelectric ring **103**) being configured to be press fit within receiving tube **411** and with annular flange **413** being configured to be press fit within engagement tube **407**. When assembled, the lower section of tubular member **102** is fully encompassed by receiving tube **411** and forms a passageway directly into aerosol chamber **414**. The bottom surface of piezoelectric ring **103** abuts the upper end of receiving tube **411** to securely hold vibration system **10** within engagement tube **407**.

(31) Referring now to FIGS. **4b** and **4c**, piezoelectric ring **103** may be supplied with an electric

current by wires **416** and **417** from batteries or another power source (not shown). Each of wires **416** and **417** may be shaped in a “c-clip” arrangement and respectively nestled into grooves **418** and **419** cut around the periphery of walls **415** of upper housing **401**. Terminal end **424** of wire **416** may enter engagement tube **407** through opening **422** in wall **415** and make electrical contact with the upper surface of piezoelectric ring **103**. Terminal end **425** of wire **417** may enter engagement tube **407** through hole **423** in wall **415** of upper housing **401** and make electrical contact with the lower surface of piezoelectric ring **103**.

(32) Center portion **110** of aperture plate **101** may be dome-shaped in geometry, although other shapes may be used. Also, center portion **110** may include apertures that taper from the rear side (facing reservoir **402**) to the front side. When aerosolizer **40** is placed in a generally vertical orientation, the liquid from reservoir **402** may be delivered to and rest on the rear side of center portion **110** by force of gravity. Piezoelectric ring **103** is configured to radially expand and contract when actuated by alternating electric fields supplied by wires **416** and **417**. In so doing, the wall of tubular member **102** also constricts and expands. In this way, center portion **110** vibrates axially so as to eject liquid droplets from its front side and out the opening in aerosol chamber **414**.

(33) One advantage of using vibration system **10** is that aerosolizer **40** may be constructed so that vibration system **10** is removable from upper housing **401** and lower housing **405**. In this way, vibration system **10** (which contains relatively expensive piezoelectric ring **103**) may be reused in other applications. Upper housing **401** and lower housing **405**, which may be able to be produced relatively inexpensively, may be discarded after use. Another advantage of using vibration system **10** is that the ends of tubular member **102** may be connected directly to rigid bodies, such as wall **406** of upper housing **401** and wall **412** of lower housing **405**, without affecting the oscillating amplitude of aperture plate **110**. This enables aerosolizer **40** to more efficiently produce liquid droplets.

(34) Referring now to FIG. **4d**, aerosolizer **40'** is similar to aerosolizer **40** and comprises upper housing **401'** containing reservoir **402'**, detachable lower housing **405'** containing aerosol chamber **414'**, and vibration system **10'**, which comprises tubular member **102'**, piezoelectric ring **103'** and aperture plate center portion **110'**. However, in aerosolizer **40'**, the upper section of tubular member **102'** is not press fit into discharge tube **404'** (as shown in FIG. **4a**), but rather O-rings **420** may be positioned to fill a gap between the upper section of tubular member **102'** and discharge tube **404'** of housing **401'**. In this way, a liquid-tight seal is formed that prevents liquid from reservoir **402'** from contacting piezoelectric ring **103'**. Similarly, O-rings **421** may be positioned to fill a gap between the lower section of tubular member **102'** and receiving tube **411'** of lower housing **405'** to form a liquid-tight seal that prevents aerosol produced from aperture plate center portion **110'** from contacting piezoelectric ring **103'**. Accordingly, piezoelectric ring **103'** may be protected from contamination that may prevent it from being reusable after removal from aerosolizer **40'**, without the tight dimensions required for a press fit, as described in connection with aerosolizer **40**. Since O-rings **420** and **421** are positioned on both top and bottom surfaces of piezoelectric ring **103'** and serve to suspend piezoelectric ring **103'** from direct contact with upper housing **401'** and lower housing **405'**, O-rings **420** and **421** may also have a dampening effect that reduces the undesirable transfer of vibration from piezoelectric ring **103'** to upper housing **401'** and lower housing **405'**.

(35) Referring now to FIGS. **5a** and **5b**, another embodiment of the present invention will be described. Vibration system **50** comprises a piezoelectric ring **502** having a central opening **504**. Piezoelectric ring **502** may be constructed of a piezoelectric material that radially expands and contracts when actuated, as previously discussed. Tubular member **506** is disposed within opening **504** and is adapted to hold aperture plate **508** within its internal lumen **509** using any of the techniques described herein. Tubular member **506** may be constructed of a rigid material, such as a hard plastic, metal, ceramic or the like. Tubular member **506** may optionally include projections **510** to provide a good mechanical contact with piezoelectric ring **502**. As an alternative, tubular member **506** may be tapered to assure a good mechanical contact.

(36) Tubular member **506** may include one or more resilient segments **511** radially extending from locations on its inner circumference to corresponding locations on its outer circumference. These segments may be constructed from an elastomeric material and positioned in various locations. Resilient segments **511** permit tubular member **506** to be constructed of a rigid material (for securely holding aperture plate **508**) while also permitting tubular member **506** to radially expand and contract with piezoelectric ring **502**. More specifically, as tubular member **506** is constricted by piezoelectric ring **502**, resilient segments **511** compress to reduce the diameter of lumen **509**. When piezoelectric ring **502** radially expands, resilient segments **511** expand to increase the diameter of lumen **509**. Hence, the amount of expansion and contraction may be varied based in part on the size, number and types of resilient materials used.

(37) Conveniently, vibration system **50** may be coupled to a reservoir of an aerosolizer (not shown) to permit a liquid to be supplied to aperture plate **508**. Also, other liquid delivery systems could be used as well, such as wicking systems, and the like. Alternatively, vibration system **50** may be incorporated into other systems, such as nebulizers, ventilators and the like.

(38) FIG. **6** illustrates vibration system **50**, as shown in FIGS. **5a** and **5b**, with an outer ring **512** disposed about the outer circumference of piezoelectric ring **502**, which in turn is disposed around tubular member **506**. Ring **512** may be employed to adjust the operating frequency of piezoelectric ring **502**. In many applications, it is desirable to operate piezoelectric ring **502** at a frequency of about 130 KHz, which is the approximate resonance frequency of the aperture plate. When piezoelectric ring **502** is constructed from a piezoceramic material, its frequency is inversely proportional to its diameter where:

$$f = \left(\frac{1}{2} \pi r \right) \sqrt{\frac{E}{\rho}}$$

(39) Hence, if the diameter of the piezoelectric ring **502** is made larger to reduce the frequency of the piezoelectric ring, the piezoelectric ring **502** may be too large for certain applications. A low operating frequency of piezoelectric ring **502** may result because the piezoelectric material is “soft” and heavy. To increase the frequency without increasing the diameter, outer ring **512** (which may be constructed of a stiff and lightweight material, such as silicon nitride) may be added. The combination of ring **512** and piezoelectric ring **502** serves to increase the frequency to the desired range.

(40) Referring to FIGS. **7a** and **7b**, an embodiment of an aerosolization system in accordance with the present invention will be described. System **80** includes an aerosolizer that, for convenience of discussion, includes vibration system **50** having ring **512**, as shown in FIG. **6**, although it will be appreciated that other vibration systems of the invention could be used as well. Coupled to (or integrally formed with) tubular member **506** of vibration system **50** is a container **802** for holding a liquid. Conveniently, a lid **804** may be provided to close container **802** after filling it with a liquid. Also coupled to tubular member **506** is an outlet **806** through which an aerosol produced by aperture plate **508** may be dispensed. O-rings or gasket seals **805** may be disposed between container **802** and vibration system **50**, and between vibration system **50** and outlet **806** to provide adequate sealing and cushioning between the components.

(41) One particular feature of aerosolization system **80** is that piezoelectric ring **502** has a large enough inner diameter that it may be slid over outlet **806** and container **802**. In this way, system **80** may be easily assembled and disassembled to remove piezoelectric ring **502**. Further, piezoelectric ring **502** does not come into contact with any liquids and therefore may be reused with another aerosolization system. Further, container **802**, tubular member **506** and aperture plate **508** may be constructed to be relatively inexpensive so that they may be disposed of following use. Also, system **80** may easily be incorporated into other systems, such as hand-held nebulizers, ventilators and the like.

(42) In operation, container **802** is filled with a liquid and lid **804** is put in place. Piezoelectric ring **502** is slid over container **802** and placed over tubular member **506**. An electric current is supplied to piezoelectric ring **502** to cause it to expand and contract. In so doing, liquid that is in contact

with aperture plate **508** is ejected as liquid droplets into outlet **806**. Following use, container **802** may be refilled, or may be discarded while saving piezoelectric ring **502**.

(43) FIG. **8** illustrates another embodiment of the invention wherein aerosolization system **90** includes tubular member **906** comprising a sharpened end **901** and a discharge end **903**. As previously described, tubular member **906** also contains an aperture plate (not shown) across its internal lumen. Piezoelectric ring **904** is disposed around tubular member **906**. When aerosolization system **90** is not in use, sharpened end **901** may have a cover (not shown) that protects it from damage and contamination. When ready for use, the cover may be removed and sharpened end **901** may be inserted through the membrane top of a vial **902**, which contains liquid **905** to be aerosolized. Liquid **905** is then delivered through sharpened end **901** and the lumen of tubular member **906** to the aperture plate contained therein. Piezoelectric ring **904** may be actuated to vibrate the aperture plate and thereby aerosolize liquid **905** in the manner previously described. The resultant aerosol is then dispensed through discharge end **903**. After use, vial **902** may be removed from sharpened end **901** and discarded, piezoelectric ring **904** may be removed from the assembly for re-use, and the remaining assembly may be discarded.

(44) As previously mentioned, the aerosolizers described herein may be incorporated into other systems. Example of ventilator systems are described, for example, in co-pending U.S. patent application Ser. No. 10/828,765, filed Apr. 20, 2004, the complete disclosure of which is herein incorporated by reference. The system described therein is particularly useful in neo-natal and infant continuous positive pressure airway pressure (CPAP) therapies. Accordingly, an aerosolizer of the present invention may be coupled to such a ventilator or CPAP circuit to supply aerosolized medicament to a patient's respiratory system, e.g., through a patient interface device. When the treatment is finished, the aerosolizer, or certain components thereof, may be removed and re-used, while other components of the system may be discarded.

(45) As another example, the aerosolizer of the present invention may be incorporated in a nebulizer such as described in co-pending U.S. patent application Ser. No. 10/833,932, filed Apr. 27, 2004, the complete disclosure of which is herein incorporated by reference. The nebulizer comprises a main housing coupled to an aerosolizer housing, which may comprise an aerosolization system such as previously described in connection with aerosolizer **40** shown in FIGS. **4a**, **4b** and **4c**, including a reservoir for holding a liquid medicament that is to be aerosolized and a vibration system according to the present invention having an aperture plate with a plurality of tapered apertures extending between a first surface and a second surface, as described in U.S. Pat. Nos. 5,164,740, 5,586,550, 5,758,637, and 6,085,740, the entire contents of which are incorporated herein by this reference. The nebulizer may also have a mouthpiece coupled to the main housing. At least a portion of the tubular member of the vibration system of the present invention may be disposed in the housing so that liquid droplets are ejected through the mouthpiece to permit a patient to inhale the aerosolized medicament. The apertures in the aperture plate may be sized to produce an aerosol in which about 70% or more of the droplets by weight have a size in the range from about 1 to about 5 micrometers. Following use, the aerosol housing may be removed from the main housing. The liquid may be refilled, or one or more components may be replaced. For example, the vibration system may be removed and reused with another nebulizer.

(46) One embodiment of the present invention provides a method of treating a patient that exhibits one or more symptoms of infection or other respiratory disease or disorder. The method generally comprises the steps of: providing a vibration system comprising a circular vibratable aperture plate having an outer circumference, a tubular member concentrically disposed about the outer circumference of the vibratable plate, wherein the tubular member has an outer circumference, and an annular vibration-inducing member concentrically disposed about the outer circumference of the tubular member, wherein the vibration-inducing member is radially expandable and contractable to cause the aperture plate to vibrate in the axial direction; supplying a liquid medicament to the vibration system; actuating the vibration-inducing member to vibrate the aperture plate and

aerosolize the medicament; and supplying the aerosol to the patient's respiratory system.

(47) An aerosol generator in accordance with the present invention has the ability to produce a high flow of aerosol relative to the power input. For example, when standard saline solution (2% NaCl) is used, the flow rate of aerosol having a volumetric median diameter (VMD) of 4 microns may be 15 microliters/sec and the power consumption of the generator may be 3 watts.

(48) The invention has now been described in detail for purposes of clarity and understanding. However, it will be appreciated that certain changes and modifications may be practiced within the scope of the appended claims.

Claims

1. An aerosol generator comprising: an upper housing member having an upstream reservoir portion configured to receive liquid for aerosolization; a lower housing member forming a downstream portion of the aerosol generator; a vibration system including: a vibratable member including a center portion having a plurality of apertures therein and an annular flange surrounding the center portion; a support member including a central hole and an annular mounting portion for receiving the annular flange of the vibratable member; a ring-shaped piezoelectric member, wherein the vibratable member, support member, and piezoelectric member are separate elements bonded together and secured between the upper housing and the lower housing members; a pair of electric lines in electrical communication with the piezoelectric member; an upper ring-shaped seal on an upstream side of the vibration system; and a lower ring-shaped seal on a downstream side of the vibration system; wherein the piezoelectric member is configured to vibrate the vibratable member in response to the pair of electric lines transmitting an electric current to the piezoelectric member; and wherein the upper and lower ring-shaped seals are configured to seal the piezoelectric member from liquid in the reservoir.
2. The aerosol generator of claim 1, wherein the ring-shaped piezoelectric member has an outer diameter that is larger than an opening in an upstream end of the lower housing member.
3. The aerosol generator of claim 2, wherein the central hole of the support member and a central opening in the piezoelectric member share a common central axis, and the central opening is larger than the central hole.
4. The aerosol generator of claim 3, wherein the center portion of the vibratable member extends in a downstream direction with respect to the annular flange.
5. The aerosol generator of claim 4, wherein the upper housing member and the lower housing member overlap one another when secured together.
6. The aerosol generator of claim 5, wherein the plurality of apertures of the vibratable member are tapered.
7. The aerosol generator of claim 6, wherein the support member is metal.
8. The aerosol generator of claim 7, wherein the vibratable member is configured to aerosolize the liquid in response to oscillation of the vibratable member.
9. The aerosol generator of claim 8, wherein the vibratable member is configured to eject the liquid as aerosol droplets via the plurality of apertures.
10. The aerosol generator of claim 9, wherein the upper housing is detachably engaged with the piezoelectric member.
11. An aerosol generator comprising: a plate including a dome-shaped center portion with one or more apertures and an outer portion surrounded by a support structure engaging the outer portion of the plate to suspend the dome-shaped center portion concentric with the support structure; a ring-shaped piezoelectric member in physical communication with the plate and configured to oscillate the plate; a pair of electric lines in electrical communication with the ring-shaped piezoelectric member; an upper housing forming a fluid reservoir for holding a fluid in fluid communication with the one or more apertures of the plate; a lower housing member forming a downstream portion

of the aerosol generator, wherein the ring-shaped piezoelectric member has an outer diameter that is larger than an opening in an upstream end of the lower housing member; and a gasket forming a seal preventing the fluid from contacting the piezoelectric member, wherein the ring-shaped piezoelectric member is configured to oscillate the dome-shaped center portion in response to the pair of electric lines transmitting an electric current to the ring-shaped piezoelectric member; wherein the plate and the support structure are disposed downstream of the fluid reservoir.

12. The aerosol generator of claim 11, wherein the plate is configured to receive the fluid along the dome-shaped center portion of the plate such that the medicament is disposed over the one or more apertures.

13. The aerosol generator of claim 12, wherein the plate is configured to aerosolize the fluid in response to the ring-shaped piezoelectric member oscillating the dome-shaped center portion.

14. The aerosol generator of claim 13, wherein the plate is configured to eject the fluid through the dome-shaped center portion as aerosol droplets via the one or more apertures.

15. The aerosol generator of claim 14, wherein the gasket comprises an elastomer seal or O-ring.

16. The aerosol generator of claim 15, wherein the plate and the support structure are surrounded by a wall that extends from the upper housing to a position below the plate and the support structure with respect to the upper housing.

17. An aerosol generator comprising: a vibratable plate including a center portion and an outer supporting portion for supporting the center portion disposed about the center portion, the center portion having a plurality of apertures, a piezoelectric member in physical communication with the vibratable plate through the outer supporting portion; a pair of electric lines in electrical communication with the piezoelectric member; an upper housing and a lower housing, the upper housing and the lower housing forming a fluid reservoir and a fluid outlet with respect to the apertures, such that a fluid can flow through the apertures to the fluid outlet from the fluid reservoir; a first gasket between the upper housing and the vibratable plate; and a second gasket between the lower housing and the vibratable plate the second gasket located downstream of the first gasket, wherein the piezoelectric member is configured to move the supporting portion to thereby oscillate the vibratable plate in response to the pair of electric lines transmitting an electric current to the piezoelectric member, such that the fluid received along the vibratable plate from the fluid reservoir is ejected as aerosolized droplets via the plurality of apertures to the fluid outlet in response to the vibratable plate oscillating, and wherein the piezoelectric member is isolated from contact with liquids.

18. The aerosol generator of claim 17, wherein the first gasket or the second gasket is an elastomer seal.

19. The aerosol generator of claim 17, wherein the first gasket is disposed on an upstream side of the vibratable plate and the second gasket is disposed on a downstream side of the vibratable plate.
